Reusable Error Notification Framework with Dynamic File Reconstruction and Attachment in Oracle Integration Cloud

This document describes a reusable Oracle Integration Cloud (OIC) integration designed to handle error events emitted by parent integrations. When an error occurs in any OIC flow, a structured JSON payload is posted to this integration's REST endpoint. The integration then decodes attachments, reconstructs the file, builds a rich HTML email body, and dispatches a formatted error notification email to the relevant recipients — optionally including the reconstructed file as an attachment.

Process Flow

Receive Error Payload (JSON)

↓

Decode Base64 Content

↓

Reconstruct File

↓

Prepare HTML Email Body (HTML)

↓

Attach File

↓

Send Notification Email

Integration Architecture

The integration follows a linear pipeline with a conditional branch for attachment handling:

1. **Receive Error Payload (REST Trigger — POST /SendNotif)**

The integration is triggered by an HTTP POST to the /SendNotif endpoint. The JSON payload carries all error metadata including RICE_ID, error message, timestamps, status, and optionally a Base64-encoded attachment file.

2. **Assign Global Variables (AssignGlobalVars)**

Initial variable assignment step. A variable VReference is initialized to an empty string. This prepares the scope for downstream file reference assignment.

3. **Switch: NOTIFICATION_ATTACHMENT = 'Y'**

Conditional branch. If the payload field NOTIFICATION_ATTACHMENT equals 'Y', the integration enters the file reconstruction path — executing Map, Stage File, and UpdateVars steps.

4. **Map (WritePayloadError) + Stage File (Opaque Schema)**

Maps the Base64-encoded content from NOTIFICATION_ATTACHMENT_FILE_REF into an opaque file structure, then stages the file using Opaque Schema.xsd / opaqueElement. The staged file reference (ICSFile) is captured in the VReference variable.

5. **Switch Route: ERROR_FLAG = 'Y' AND NOTIFICATION_ATTACHMENT = 'Y'**

Second routing decision. Confirms both conditions before triggering the notification action with attachment.

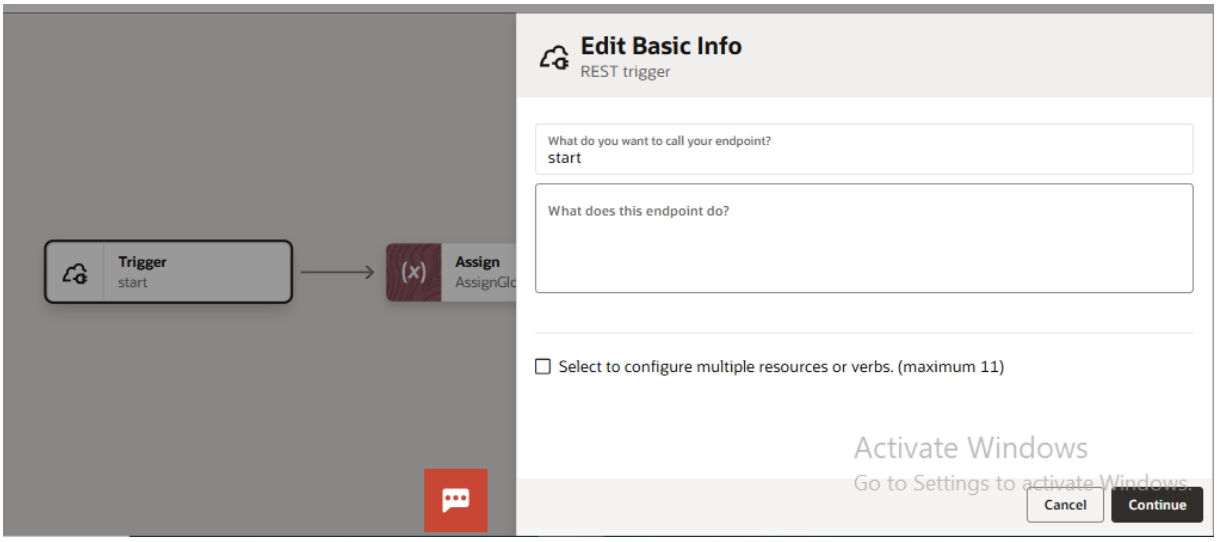
6. **Notification Action (ErrorWithAttachment)**

Sends the HTML-formatted email. To, From, Subject (concat of RICE_ID + INTEGRATION_STATUS), and full HTML body are configured. The staged file is attached to the email.

OIC Flow Configuration Screenshots

1. **REST Trigger Configuration**

The integration exposes a REST endpoint with URI /SendNotif, and HTTP method POST.



Edit Resource Configuration

REST trigger

What is the endpoint's relative resource URI?

/SendNotif

What action do you want to perform on the endpoint?

POST

Based on your selections, you can add parameters or configure a request and/or response for this endpoint.

Select any options that you want to configure:

- [ ] Add and review parameters for this endpoint
- [x] Configure a request payload for this endpoint
- [ ] Configure this endpoint to receive the response

Activate Windows
Go to Settings to activate Windows.

Cancel | Go back | Continue

Edit Request

REST trigger

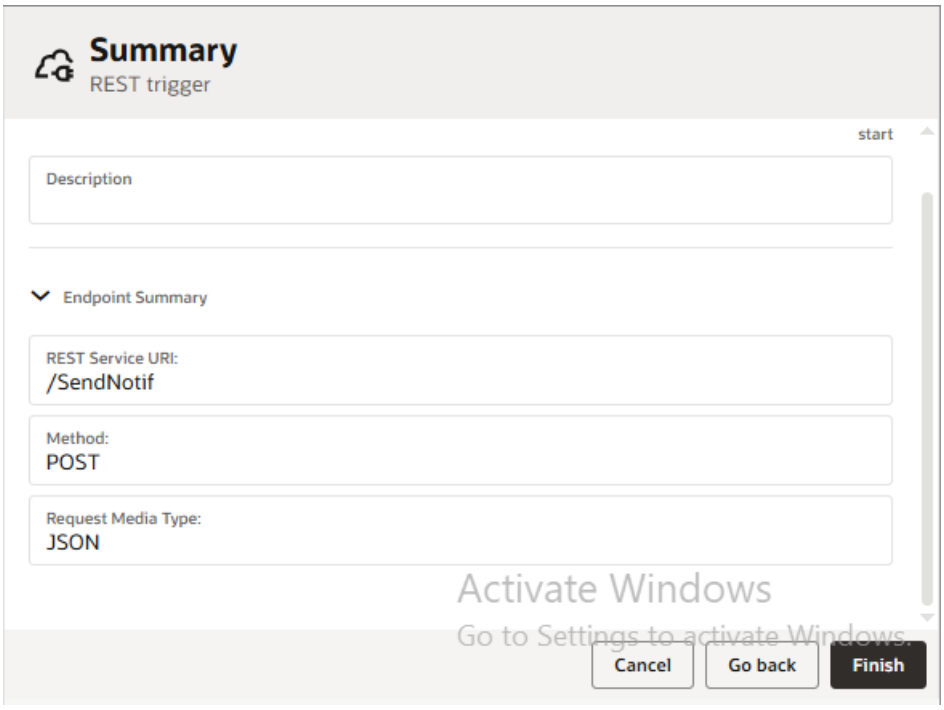
Enter Sample JSON


```
{
 "OIC_PARENT_INSTANCE_ID" : "AZERTY81726354QyxSrU",
 "RICE_ID" : "INT009",
 "OIC_START_TIME" : "2022-08-02T08:38:02.388Z",
 "OIC_END_TIME" : "2022-08-02T08:50:01.167Z",
 "NOTIFICATION_ATTACHMENT" : "Y",
 "NOTIFICATION_ATTACHMENT_FILE_REF" : "hcgjqwgcjqeg",
 "NOTIFICATION_ATTACHMENT_FILE_NAME" : "NotificationAttachmentFile",
 "ERROR_FLAG" : "Y",
 "INTEGRATION_STATUS" : "ERROR",
 "ERROR_MESSAGE" :
"Error+while+translating.%0ATranslation+exception.%0AError+occurred+while+transl
```


Cancel | Continue

Activate Windows
Go to Settings to activate Windows.

Cancel | Go back | Continue

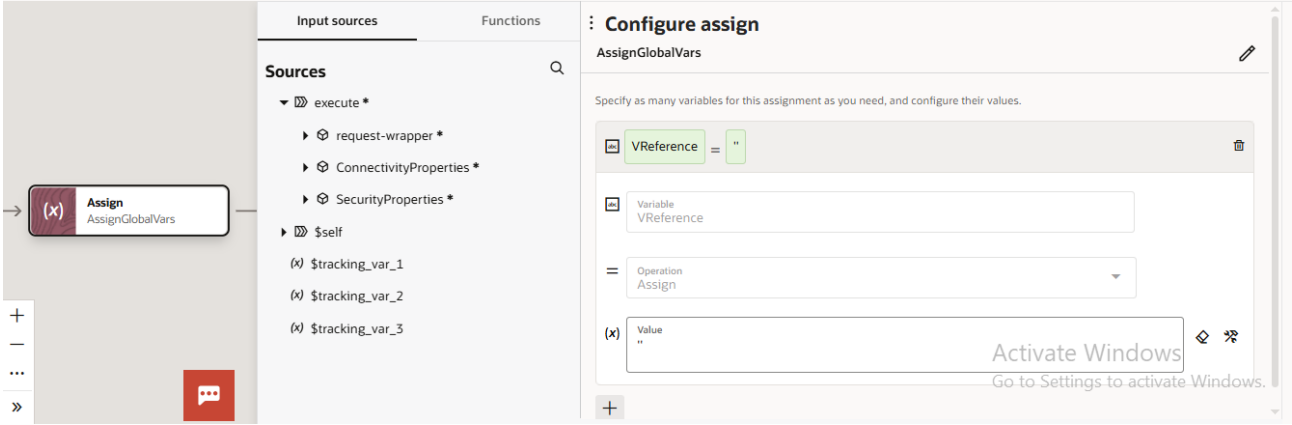


Sample JSON Payload:


```
{
  "RICE_ID": "F0xRiv3r",
  "OIC_PARENT_INSTANCE_ID": "AZERTY81726354QyxSrU",
  "ATTRIBUTE1": "System Error",
  "OIC_START_TIME": "2022-08-02T08:38:02.388Z",
  "ERROR_MESSAGE":
"Error+while+translating.%0ATranslation+exception.%0AError+occurr
ed+while+translating+content+from+file+%2Ftmp%2Ffileftp%2Ftemp_ft
p%2FReadFile_REQUESTPortType",
  "NOTIFICATION_ATTACHMENT": "Y",
  "INTEGRATION_STATUS": "ERROR",
  "OIC_END_TIME": "2022-08-02T08:50:01.167Z",
  "NOTIFICATION_ATTACHMENT_FILE_REF":
"YXR0cmlidXRlMXxhdHRyaWJ1dGUyfGF0dHJpYnV0ZTMKQVpFUlRZfENoaWxkfFBh
cmVudApGdXNpb258QUJDfEFDQg==",
  "NOTIFICATION_ATTACHMENT_FILE_NAME":
"NotificationAttachmentFile",
  "ERROR_FLAG": "Y"
}
```

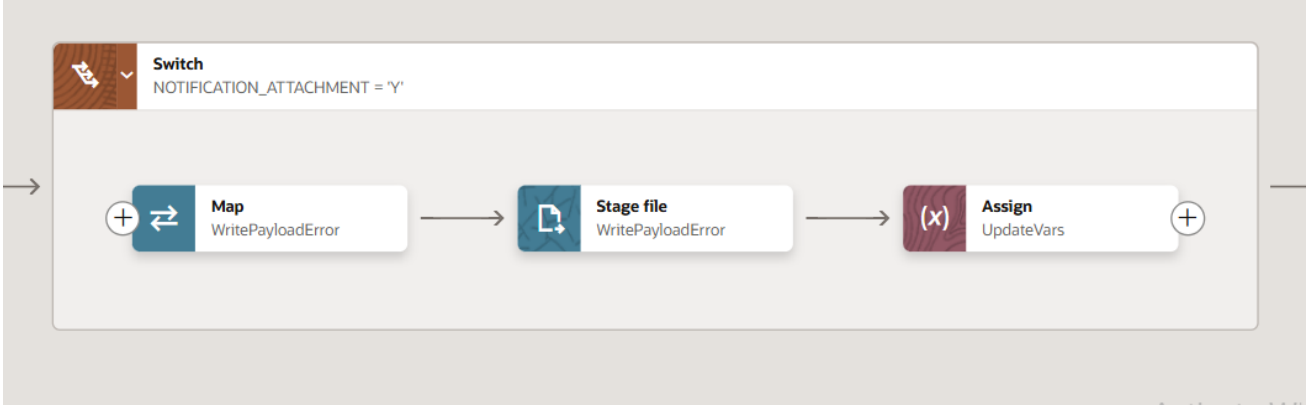

2. Assign Global Variables

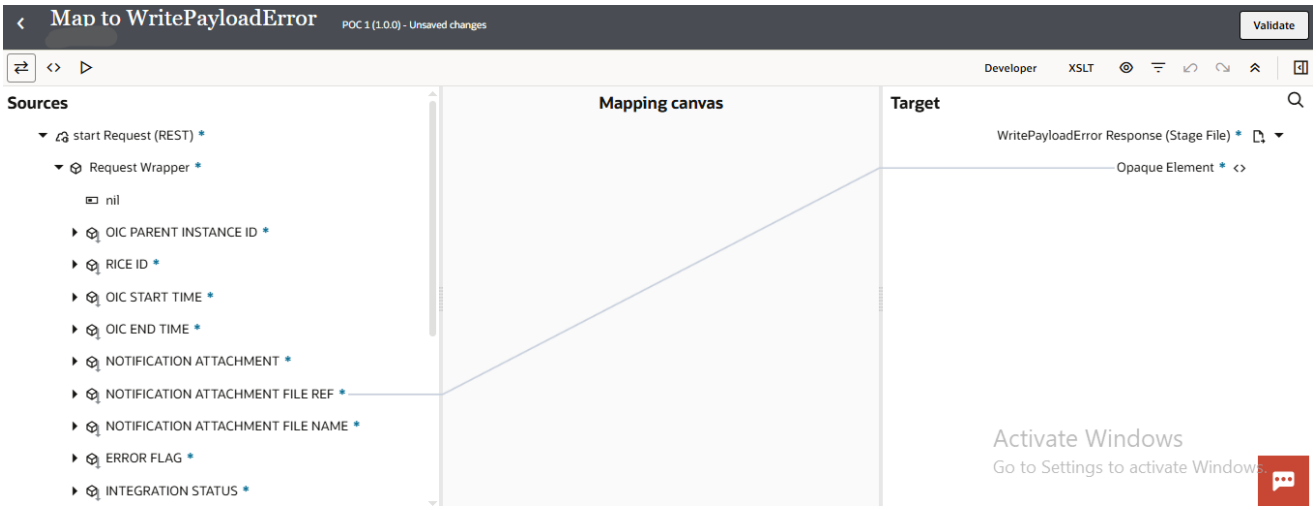
The AssignGlobalVars step initializes the VReference variable to an empty string before any file staging occurs.



3. Switch — Attachment Conditional Branch

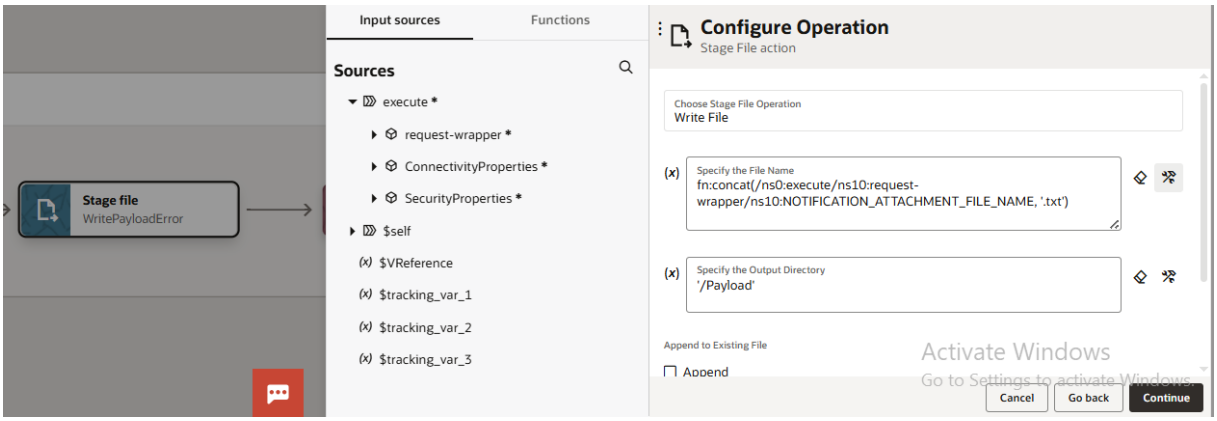
The Switch action checks whether NOTIFICATION_ATTACHMENT = 'Y'. If true, the flow proceeds through Map, Stage File, and variable update steps.

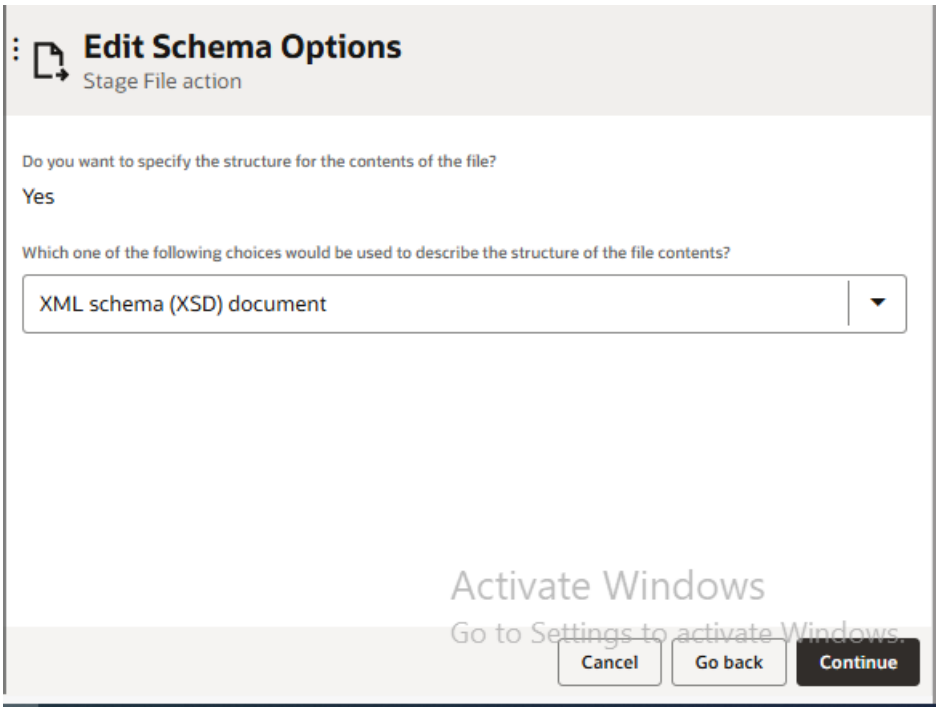




3.1 Stage File — Opaque Schema

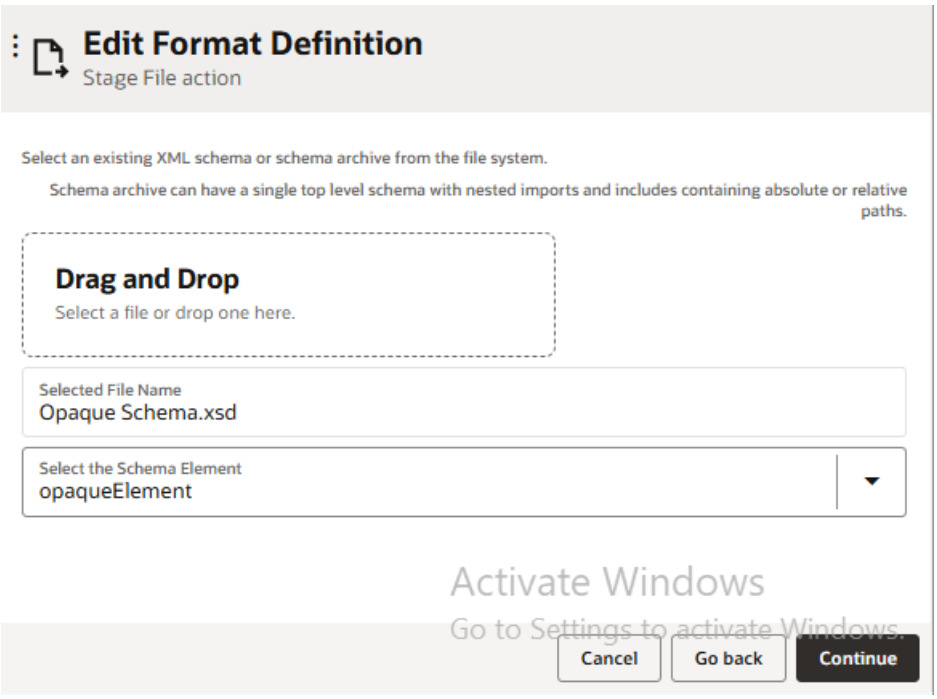
The Stage File action uses Opaque Schema.xsd with the opaqueElement schema element to reconstruct the binary/text file from the Base64-encoded payload field.





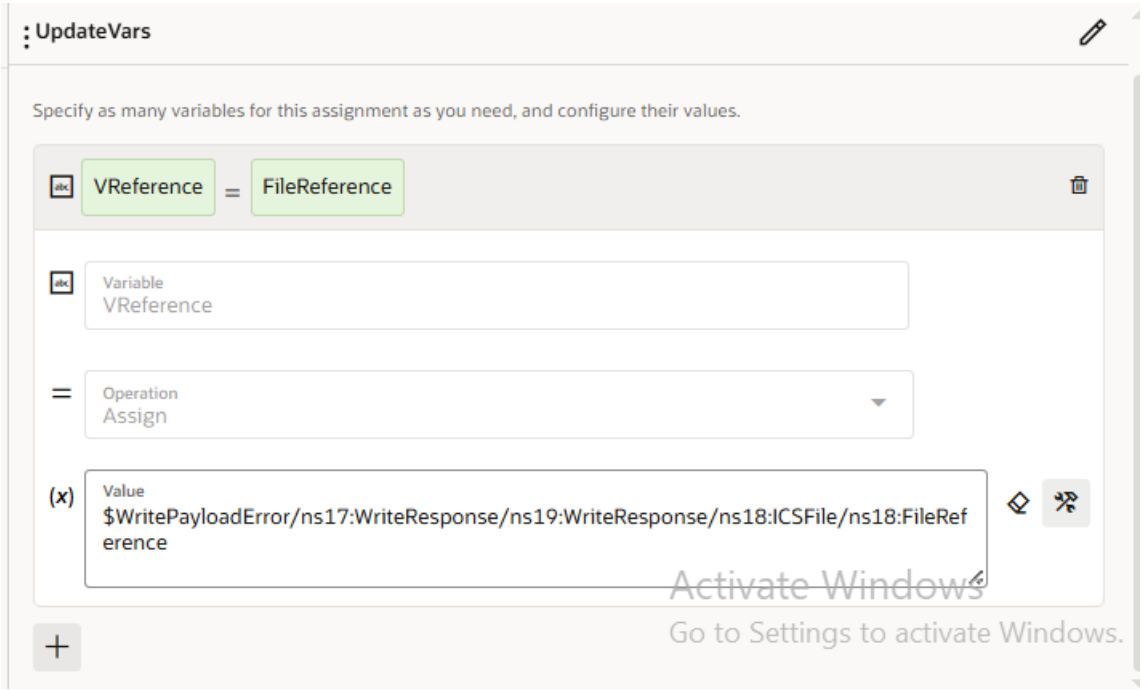
Opaque Schema.xsd

```
<?xml version = '1.0' encoding = 'UTF-8'?>
<schema
targetNamespace="http://xmlns.oracle.com/pcbpel/adapter/opaque/"
            xmlns="http://www.w3.org/2001/XMLSchema" >
      <element name="opaqueElement" type="base64Binary" />
    </schema>
```



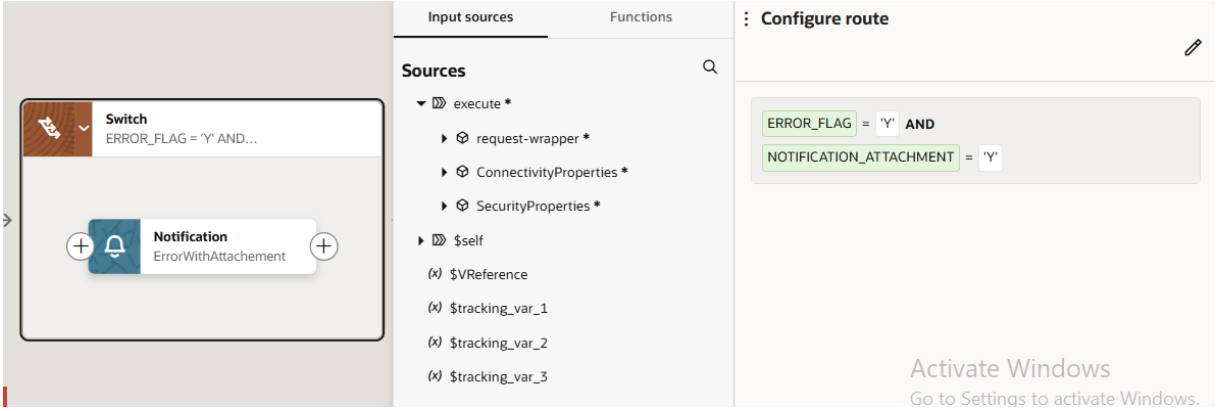
3.2 UpdateVars — Capturing the File Reference

After staging, the UpdateVars assign step captures the ICSFile reference from the WritePayloadError response into the VReference variable for later use by the notification action.



4. Route: ERROR_FLAG + NOTIFICATION_ATTACHMENT

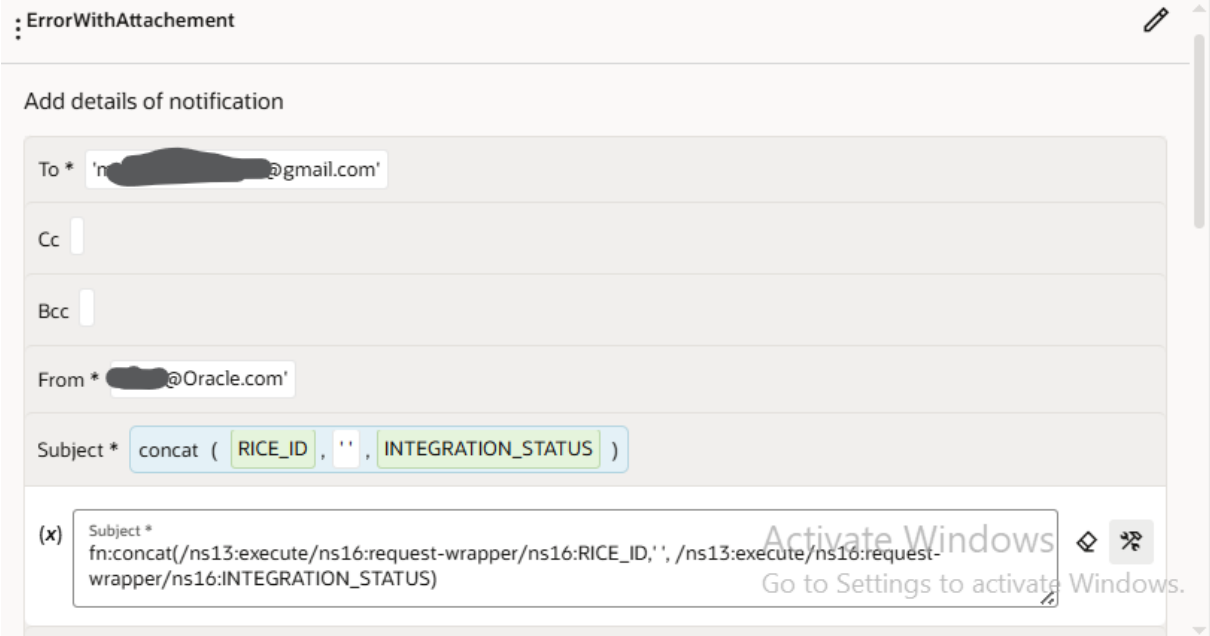
The route condition ensures the notification email is only sent when both ERROR_FLAG = 'Y' AND NOTIFICATION_ATTACHMENT = 'Y' are true simultaneously.



5. Notification Action Configuration

The ErrorWithAttachment notification action is configured with:

- To: '**********@gmail.com'
- From: '**********@Oracle.com'
- Subject: concat(RICE_ID, ' ', INTEGRATION_STATUS)
- Body: Full HTML template with dynamic OIC variables



6. HTML Email Body

The notification body contains a rich HTML template with inline CSS, two data tables (integration context and error details), and a direct OIC instance link.

Full HTML Template:

```
<html>
<head>
<style>
table {
  font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
  font-size: 13.5px;
  border-collapse: collapse;
  width: 100%;
}

p {
  font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
  font-size: 13px;
}

span {
  font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
  font-size: 13px;
}

td,
th {
  border: 1px solid #cccccc;
  text-align: left;
  padding: 10px;
  font-size: 13px;
}

th {
  background-color: #002060;;
  color: #ffffff;
  width: 30%;
}
```

```
</style>
</head>
<body>
<span>Hello,</span>
<br>
<br>
<span>The integration process encountered an error during
execution. Please review the details below:</span>
<br>
<br>
<table>
<tr>
<th>OIC Environment Name</th>
<td>{P_ENV}</td>
</tr>
<tr>
<th>OIC Instance Parent ID</th>
<td>{P_OIC_PARENT_INSTANCE_ID}</td>
</tr>
<tr>
<th>OIC Execution Start Time</th>
<td>{P_OIC_Start_Time}</td>
</tr>
<tr>
<th>OIC Failure Time</th>
<td>{P_OIC_End_Time}</td>
</tr>
<tr>
<th> Integration Status</th>
<td>{P_INTEGRATION_STATUS}</td>
</tr>
</table>
<br>
<br>
<table>
<tr>
<th>ERROR_TYPE</th>
<th>ERROR_MESSAGE</th>
```

```
</tr>
<tr>
<td>{P_ERROR_TYPE}</td>
<td>{P_ERROR_MESSAGE}</td>
</tr>
</table>
<br>
<br>
<p style="color: #000000; font-size: 14px; font-weight: normal;
font-family: Arial, Helvetica, sans-serif;"><strong>OIC Instance
Access</strong></p>
<p style="color: #000000; font-size: 14px; font-weight: normal;
font-family: Arial, Helvetica, sans-serif;">View the OIC instance
execution in one click: <a
href={P_OIC_URL}><i>OIC_INSTANCE_URL</i></a></p>
<br>
<br>
<span>**This is an automatically generated message from the OIC
Integration Cloud service. Please do not reply to this
email.**</span>
</body>
</html>
```

Body *

```
<html> <head> <style> table { font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif; font-size: 13.5px; border-collapse: collapse; width: 100%; } p { font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif; font-size: 13px; } span { font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif; font-size: 13px; } td, th { border: 1px solid #cccccc; text-align: left; padding: 10px; font-size: 13px; } th { background-color: #002060;; color: #ffffff; width: 30%; } </style> </head> <body> <span>Hello,</span> <br> <br> <span>The integration process encountered an error during execution. Please review the details below:</span> <br> <br> <table> <tr> <th>OIC Environment Name</th> <td>{P_ENV}</td> </tr> <tr> <th>OIC Instance Parent ID</th> <td> {P_OIC_PARENT_INSTANCE_ID}</td> </tr> <tr> <th>OIC Execution Start Time</th> <td>{P_OIC_Start_Time} </td> </tr> <tr> <th>OIC Failure Time</th> <td>{P_OIC_End_Time}</td> </tr> <tr> <th> Integration Status</th> <td>{P_INTEGRATION_STATUS}</td> </tr> </table> <br> <br> <table> <tr> <th>ERROR_TYPE</th> <th>ERROR_MESSAGE</th> </tr> <tr> <td>{P_ERROR_TYPE}</td> <td> {P_ERROR_MESSAGE}</td> </tr> </table> <br> <br> <p style="color: #000000; font-size: 14px; font-weight: normal; font-family: Arial, Helvetica, sans-serif;"><strong>OIC Instance Access</strong></p> <p style="color: #000000; font-size: 14px; font-weight: normal; font-family: Arial, Helvetica, sans-serif;">View the OIC instance execution in one click: <a href={P_OIC_URL}><i>OIC_INSTANCE_URL</i></a></p> <br> <br> <span>**This is an automatically generated message from the OIC Integration Cloud service. Please do not reply to this email.**</span> </body> </html>
```

Parameters

P_ERROR_MESSAGE get-content-as-string (ERROR_MESSAGE)

Parameter name
P_ERROR_MESSAGE

Description

(x) Parameter value
oraext:get-content-as-string(/ns13:execute/ns16:request-wrapper/ns16:ERROR_MESSAGE)

Parameters

P_ERROR_MESSAGE get-content-as-string (ERROR_MESSAGE)

P_OIC_Start_Time OIC_START_TIME

P_ERROR_TYPE ATTRIBUTE1

P_OIC_URL baseURL

P_INTEGRATION_STATUS INTEGRATION_STATUS

P_OIC_PARENT_INSTANCE_ID OIC_PARENT_INSTANCE_ID

P_OIC_End_Time OIC_END_TIME

P_ENV serviceInstanceName

Activate Windows
Go to Settings to activate Windows.

Attachments

VReference

(x) Value
$VReference

Activate Windows
Go to Settings to activate Windows.

The following screenshots illustrate the OIC integration execution and the notification email received upon error detection

Text File


```
{
  "RICE_ID": "F0xRiv3r",
  "OIC_PARENT_INSTANCE_ID": "AZERTY81726354QyxSrU",
  "ATTRIBUTE1": "System Error",
  "OIC_START_TIME": "2022-08-02T08:38:02.388Z",
  "ERROR_MESSAGE":
"Error+while+translating.%0ATranslation+exception.%0AError+occurred+while+translating+content+from+file+%2Ftmp%2Ffileftp%2Ftemp_ftp%2FReadFile_REQUESTPortType",
  "NOTIFICATION_ATTACHMENT": "Y",
  "INTEGRATION_STATUS": "ERROR",
```


Response

Status: 202 Accepted

Instance ID: UCktj' lqEg

Body Headers

Tracing level: **Debug**

Sat Apr 04 09:10:23.335 PM WEST 2026,
Wire Message dequeued and processed by Trigger start

09:10:23.545 PM
Message started processing

09:10:23.577 PM
Message received by Assignment AssignGlobalVars

09:10:23.578 PM
Message entered Switch

09:10:23.578 PM
Message routed through Switch Route with condition NOTIFICATION_ATTACHMENT = 'Y'

09:10:23.578 PM, 3ms
Message received by Data Mapper

09:10:23.581 PM
Data Mapping completed with Message

Activate Windows
Go to Settings to activate Windows.

F0xRiv3r ERROR Inbox ×

[redacted] Unsubscribe 7:10 PM (2 hours ago)

to me

Hello,

The integration process encountered an error during execution. Please review the details below:

OIC Environment Name	oi[redacted] 1-st
OIC Instance Parent ID	AZERTY81726354QyxSrU
OIC Execution Start Time	2022-08-02T08:38:02.388Z
OIC Failure Time	2022-08-02T08:50:01.167Z
Integration Status	ERROR

ERROR_TYPE	ERROR_MESSAGE
System Error	Error+while+translating.%0ATranslation+exception.%0AError+occurred+while+translating+content+from+file+%2Ftmp%2Ffileftp%2Ftemp_ftp%2FReadFile_REQUESTPortType

Activate Windows
Go to Settings to activate Windows

OIC Instance Access

ERROR_TYPE	ERROR_MESSAGE
System Error	Error+while+translating.%0ATranslation+exception.%0AError+occurred+while+translating+content+from+file+%2Ftmp%2Ffileftp%2Ftemp_ftp%2FReadFile_REQUESTPortType

OIC Instance Access

View the OIC instance execution in one click: *OIC_INSTANCE_URL*

This is an automatically generated message from the OIC Integration Cloud service. Please do not reply to this email.

One attachment • Scanned by Gmail ⓘ Add to Drive

NotificationAttac...

Reply Forward

Activate Windows
Go to Settings to activate Windows.

NotificationAttachmentFile.txt Open with Google Docs

```
attribute1|attribute2|attribute3
AZERTY|Child|Parent
Fusion|ABC|ACB
```

Activate Windows
Go to Settings to activate Windows.